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I am an economist who studies the valuation and allocation of public and environmental resources. Mechanisms and valuations always rest on assumptions: zero transaction cost, constant biological stocks, and strategic behavior are three that are relevant to my own research. I study the resilience of those mechanisms and the value of these environmental resources when these assumptions are relaxed. I pursue these questions using economic theory, structural welfare estimation, and reduced-form causal inference. My dissertation studies carbon offset markets and outdoor recreation, but my interests span environmental economics and mechanism design more broadly.

Asymmetric information and costly participation in voluntary conservation markets (Job market paper). Carbon offset markets are widely treated as, at worst, harmless—a device that lowers firms’ compliance costs and can only improve welfare. My job market paper shows that this need not hold: once participation is costly and information is asymmetric, offset markets can reduce welfare rather than improve it. Offset markets allow emissions-regulated firms to meet their compliance obligations by paying for abatement via avoided deforestation, improved agricultural practices, and other nature-based activities. The standard design assigns each participant a baseline and pays for abatement above it, and nearly every theoretical treatment assumes this trade is frictionless. Extending the standard model to allow transaction costs overturns three conclusions the literature has widely held: that offsets are innocuous at worst, that raising the baseline improves efficiency, and that the optimal offset price is never below the Pigouvian level.

Until now, the literature has agreed that welfare is increasing in the baseline; the intuition is that requiring agents to provide more abatement upfront leads to a lower average cost for each unit of abatement. This is reversed when transaction costs are present, as higher baselines tend to select agents with higher costs for participation (which creates less—potentially negative—welfare). The literature has also agreed that the price of an offset does not affect the overall “additionality” of the market, and that the optimal price in such a market will never be less than the Pigouvian standard. This is not always true when participating requires a transaction cost, and I highlight a case where a sub-Pigouvian price creates higher additionality and welfare—dominates the market at the Pigouvian price, even when this baseline is tailored for the Pigouvian price.

The largest offset programs in the world combine unobservable counterfactuals with heterogeneous transaction costs, and the response in the policy sphere has focused on measuring and ensuring additionality. Such advances genuinely improve the regulator’s ability to make policy, but they leave the transaction costs—which can be independent of additionality—unaddressed. My paper characterizes when a market is distorted enough that it should be substantially reformed (or even shut down). In doing so, I provide three novel results on the value of offset markets, the baseline’s effect on efficiency, and the impact of pricing below the Pigouvian standard.

Transaction costs of emissions offsets: Evidence from California (Under review). This paper supplied the original empirical motivation for the theory in my job market paper. High-emitting California firms must surrender permits or offsets against all of their emissions. For the first decade of the program,

offsets were priced roughly 30% below permits, yet many firms—in some years, a majority—used no offsets at all. Taken at face value, this looks like money being left on the table. But the transaction costs required to use offsets rationalize this behavior; previous literature has identified similar costs in the European Union’s emissions trading system. I provide extensive descriptive evidence from raw firm-level compliance data to illustrate the seeming mismatch between cost-minimization and observed offset use, and I use econometric methods to estimate the magnitude of these transaction costs.

The identification exploits a feature of California’s regulatory policy: the share of compliance that a firm can meet with offsets is capped, and that cap bounds the transaction costs from both sides. The intuition is this: if a firm uses offsets, its transaction cost must be smaller than its cost savings; likewise, if a firm does not use offsets, its transaction cost must be greater than the maximum potential cost savings it could have earned by using them. Building empirical strategies out of institutional details like this cap is characteristic of how I approach empirical work.

Outdoor recreation and nonmarket valuation. Nonmarket valuation drew me into economic research, and it remains an active part of my agenda. My third dissertation chapter, “The impacts of climate-change induced fish migration on angler welfare,” pairs standard nonmarket valuation tools with fish population models drawn from the natural science literature under several climate scenarios. Linking data on shoreline angler trips along the east coast with data on fish stocks and expected climate-dependent future fish stocks lets me trace an indirect channel from sea temperature to fish stocks to catch rates to recreation behavior. This endogenizes fish stocks, a quantity usually held fixed, but one that may substantially shift the value of recreation, particularly in areas further south or north. In related co-authored work, we use a two-decade panel of shoreline anglers (2004–2024) to quantify bimonthly userday values at over 2,600 sites across the coast and apply internal meta-analysis to measure the stability of these values across time and space.

Impacts of centralized budget policy on national parks (PERC graduate fellowship, Summer 2025). In this paper, I examine whether a 2004 policy that gave park managers greater control over user-fee revenue crowded out central federal appropriations. To do this, I digitized three decades of National Park Service budget documents, assembling a panel of more than 10,000 park-year observations spanning 1986–2019. Comparing fee-charging parks with non-fee-charging parks through difference-in-differences, trend-break, and synthetic-control designs, I cannot reject the null hypothesis that no systematic crowding out occurred. This result contributes to a larger discussion about local and federal regimes, suggesting that the policy added genuine funding rather than merely reshuffling dollars. This project combines my research interest in public goods mechanisms with the original data creation that I enjoy.

A matching mechanism in the field (work in progress). In a newer co-authored project, topically distinct from my dissertation but emblematic of the kind of research I like to do, I study NC State’s centralized engineering major-assignment system, which closely resembles the “Boston mechanism” from matching theory. A common critique of these mechanisms is that they are not strategyproof: students may manipulate their outcome by strategically misreporting their preferences. My co-author and I surveyed engineering students to elicit their preferences when they first came to NC State, and at the time they submitted their preferences to the major-assignment mechanism. We find that (i) more than 95% of

students submitted their true preferences to the mechanism, and (ii) about 65% of students changed their minds on what their top-choice major was after the first year. (i) is a somewhat unexpected result, and it motivates the theory we then build: under mild conditions about student preference formation—which are defended using (ii)—the Boston mechanism yields equivalent outcomes to a strategyproof mechanism. This project joins two aspects of research that I consider representative of my future work: collecting original data, and explaining outcomes that appear puzzling at first.

Agenda. What ties these projects together is the distance between how mechanisms and valuations are designed to perform and how they perform in practice or once these assumptions are relaxed. Going forward, I plan to continue working at the nexus of environmental economics and mechanism design by creating original datasets to study real-world policies or conduct nonmarket valuation, and to use economic theory when reality appears different from prediction.